Week 5 - Written Results: Tasks 3, 4 and 5

Neural network experiments on the Breast Cancer and Digits datasets (TensorFlow/Keras). Full code and training logs are in the accompanying notebooks.

Task 3 - Training Graphs and Explanation (Breast Cancer ANN)

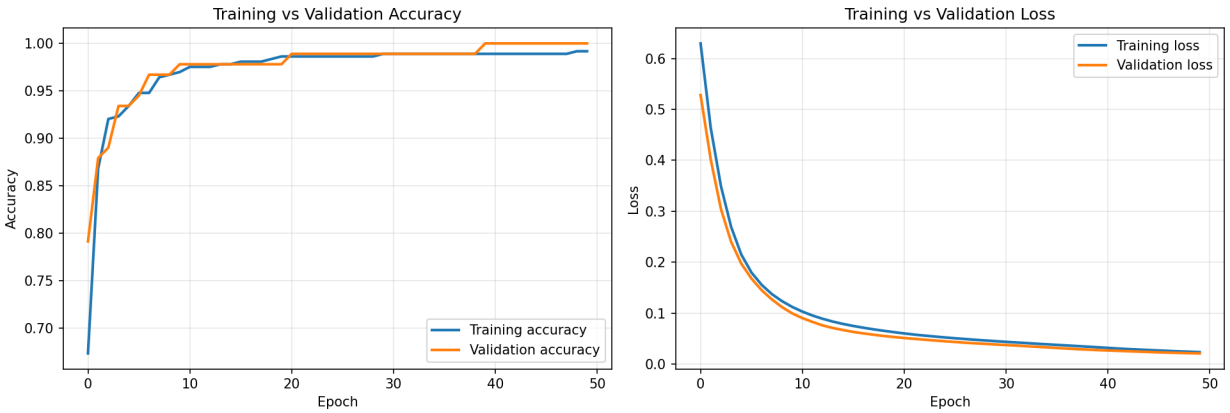


Figure 4: Training vs validation accuracy (left) and loss (right) over 50 epochs.

During training, accuracy rose quickly. Training accuracy climbed from 67.3% in epoch 1 to 99.2% in epoch 50, while validation accuracy rose from 79.1% to 100%. Loss moved the opposite way: training loss fell from 0.630 to 0.023 and validation loss from 0.528 to 0.021, with the steepest drop in the first 10 to 15 epochs. The model is clearly improving, and the flattening curves show it is close to convergence. I see no strong sign of overfitting, because validation loss is still falling, tracks training loss closely, and validation accuracy is not below training accuracy. Overfitting would appear as training loss continuing to fall while validation loss rises and the accuracy gap widens. One caution: the validation set holds only 91 samples, so 100% validation accuracy is optimistic. The test accuracy of 95.6% is the more honest figure. EarlyStopping or Dropout would be sensible safeguards, and training for more epochs is unlikely to help much because the curves have already levelled off.

Task 4 - Test Evaluation (Breast Cancer ANN)

Test accuracy: 95.61% (test loss 0.0987) on 114 unseen samples. Malignant: precision 0.91, recall 0.98. Benign: precision 0.99, recall 0.94.

Actual	Predicted	P(Benign)	Correct?
Malignant	Malignant	0.000	Yes
Benign	Benign	1.000	Yes
Malignant	Malignant	0.001	Yes
Benign	Malignant	0.381	No
Malignant	Malignant	0.000	Yes
Benign	Benign	0.998	Yes
Benign	Benign	1.000	Yes

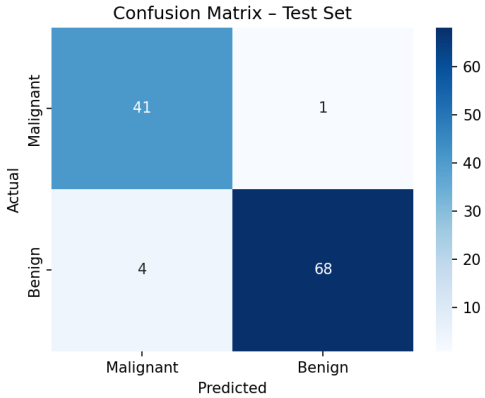


Figure 5: Confusion matrix for the test set.

The model made 5 errors on 114 test samples (95.6% accuracy). Four were benign tumours predicted as malignant, with P(benign) between 0.29 and 0.49, so they were borderline cases near the decision threshold; these are false

alarms that would lead to extra tests. One was a malignant tumour predicted as benign with high confidence (P(benign) = 0.986). This is the most serious type of error, because a real cancer would be missed. Such cases usually have measurements (radius, texture, concavity) that resemble benign tumours. Malignant recall is 0.98, which is good, but a real system should lower the threshold to reduce missed cancers and validate with cross-validation and more data.

Task 5 - Mini Project: Handwritten Digit Recognition

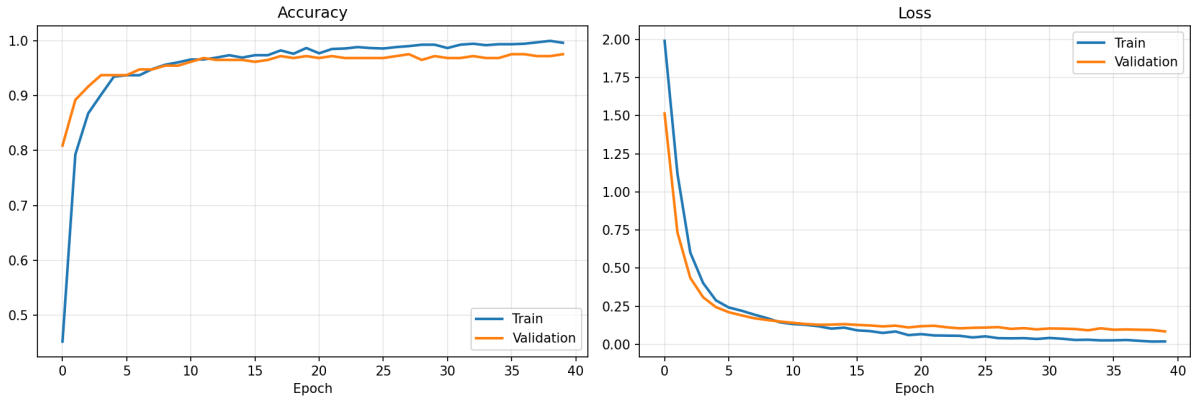


Figure 6: Accuracy and loss curves (Digits dataset, 40 epochs).

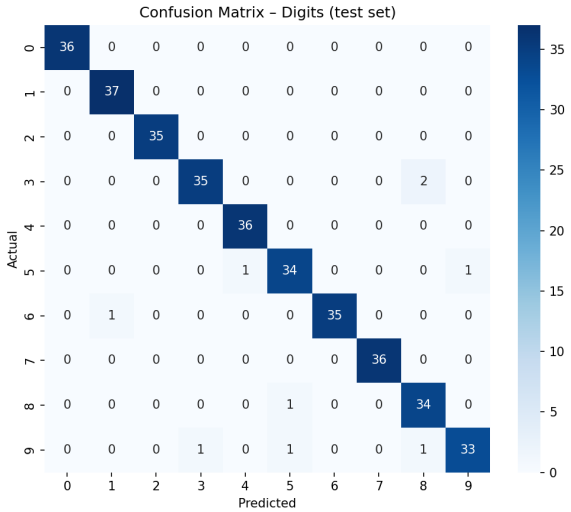


Figure 7: Confusion matrix (360 test images, 97.5% accuracy).

Problem. Recognise handwritten digits (0-9) automatically, as used in postal-code reading and form digitisation.

Dataset. The scikit-learn Digits dataset has 1,797 grayscale 8x8 images (64 pixel features) in 10 balanced classes. Pixels were scaled to 0-1 and the data split 80/20 (1,437 train, 360 test, stratified).

Architecture. A fully connected network: 64 inputs, Dense(128, ReLU), Dropout(0.2), Dense(64, ReLU) and Dense(10, Softmax). It was compiled with Adam and sparse categorical cross-entropy.

Training. The model trained for 40 epochs with batch size 32 and 20% of the training data used for validation. Training accuracy rose from 45.2% to 99.7% and validation accuracy from 80.9% to 97.6%, while validation loss fell from 1.51 to 0.085. The two curves stayed close, so overfitting was mild; Dropout helped.

Results. Test accuracy was 97.5% (9 of 360 images misclassified). Most digit classes scored an F1 of 0.96 or higher; digit 8 was weakest, and errors involved visually similar digits such as 9, 3 and 5.

Limitations. The images are tiny (8x8) and clean, a dense network ignores the 2-D structure of images, and only one train/test split was used, so the estimate has some variance.

Future improvements. Use a Convolutional Neural Network on full-size MNIST, add data augmentation, tune hyperparameters, use cross-validation and EarlyStopping, and deploy the model in a small web app. Overall, the project shows that even a small neural network can learn useful patterns directly from raw pixels, and that careful preprocessing, validation curves and a confusion matrix are what make the results trustworthy. The same workflow can be reused for other classification problems such as medical or student-performance data.